SETTING UP NETWORKING SIMULATION TOOLS AND INSTALLING WIRESHARK FOR NETWORK TRAFFIC ANALYSIS

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ABSTRACT

Network traffic analysis is a fundamental skill in cybersecurity, network administration, and digital forensics. This project focused on the installation, configuration, and utilization of Wireshark for capturing and analyzing network traffic within a home network environment. The project involved identifying active network interfaces, capturing live traffic, examining common network protocols, verifying network connectivity, and documenting implemented security controls. The analysis revealed the presence of DNS, TCP, ICMP, UDP, and TLS communications, demonstrating the practical application of packet analysis in monitoring and troubleshooting network activities. The project provided hands-on experience with network monitoring techniques commonly used by cybersecurity professionals, network administrators, and security analysts.

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1. INTRODUCTION

Modern computer networks form the backbone of communication systems used by individuals, businesses, and governments. As network usage continues to increase, cybersecurity professionals require tools that enable them to monitor, analyze, and secure network communications.

Wireshark is one of the most widely used network protocol analyzers in the world. It allows users to capture, inspect, and analyze packets traversing a network in real time. By examining captured traffic, security analysts can identify communication patterns, troubleshoot connectivity issues, investigate incidents, and detect suspicious activities.

This project focused on establishing a practical network monitoring environment using Wireshark. The project included software installation, network traffic capture, protocol analysis, connectivity testing, and documentation of implemented security controls.

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2. PROJECT OVERVIEW

This project introduced the concepts of network simulation and packet analysis through the installation and use of Wireshark.

The project aimed to provide practical experience in:

- Network monitoring
- Packet capture
- Protocol analysis
- Connectivity verification
- Security assessment
- Troubleshooting network communications

The project was conducted within a home wireless network environment.

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3. PROJECT OBJECTIVES

The objectives of this project were to:

1. Install and configure Wireshark.
2. Install and configure Npcap packet capture drivers.
3. Identify active network interfaces.
4. Capture live network traffic.
5. Analyze common network protocols.
6. Verify connectivity between devices.
7. Document network architecture.
8. Examine implemented security controls.
9. Gain practical experience with cybersecurity monitoring tools.

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4. PROJECT SCOPE

This project falls under the area of Network Monitoring and Traffic Analysis.

The scope includes:

- Packet capture
- Protocol identification
- Network connectivity testing
- Host-based security verification
- Wireless security assessment
- Network troubleshooting

The project does not involve offensive security testing or penetration testing activities.

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5. TOOLS AND SOFTWARE USED

Table 1: Tools Used

Tool: Wireshark 4.6.6 Purpose: Packet Capture and Analysis

Tool: Npcap Purpose: Packet Capture Driver

Tool: Windows 11 Purpose: Host Operating System

Tool: Wi-Fi Network Adapter Purpose: Network Traffic Source

Tool: Android Smartphone Purpose: Additional Network Device

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6. ASSETS, THREATS, AND VULNERABILITIES

6.1 Assets Identified

The following assets were identified:

• Windows Laptop • Home Router • Android Smartphone • Network Traffic • Internet Connection • Wireshark Capture Files

6.2 Potential Threats

• Packet Sniffing • Unauthorized Access • Malware Communications • Data Interception • Network Reconnaissance

6.3 Potential Vulnerabilities

• Weak Wireless Security • Unmonitored Network Traffic • Misconfigured Systems • Unauthorized Devices

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7. WIRESHARK INSTALLATION AND CONFIGURATION

Wireshark was downloaded from the official Wireshark website and installed successfully.

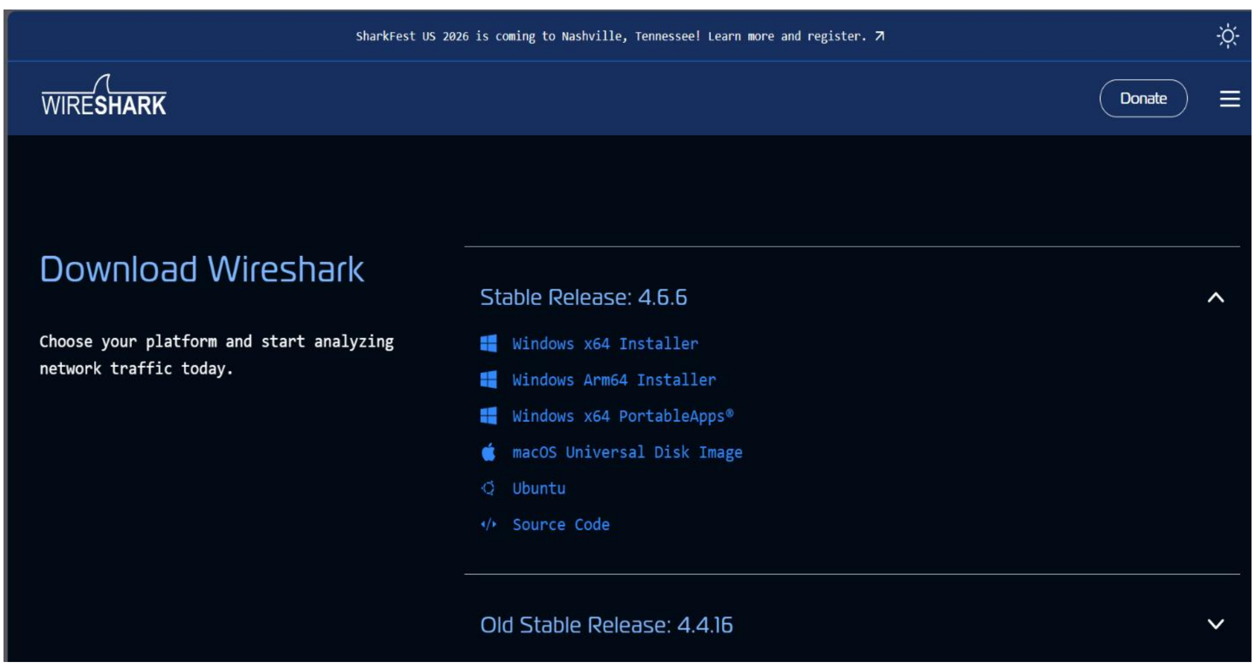
The installation included:

• Wireshark Network Analyzer • Npcap Packet Capture Driver • Protocol Analysis Components

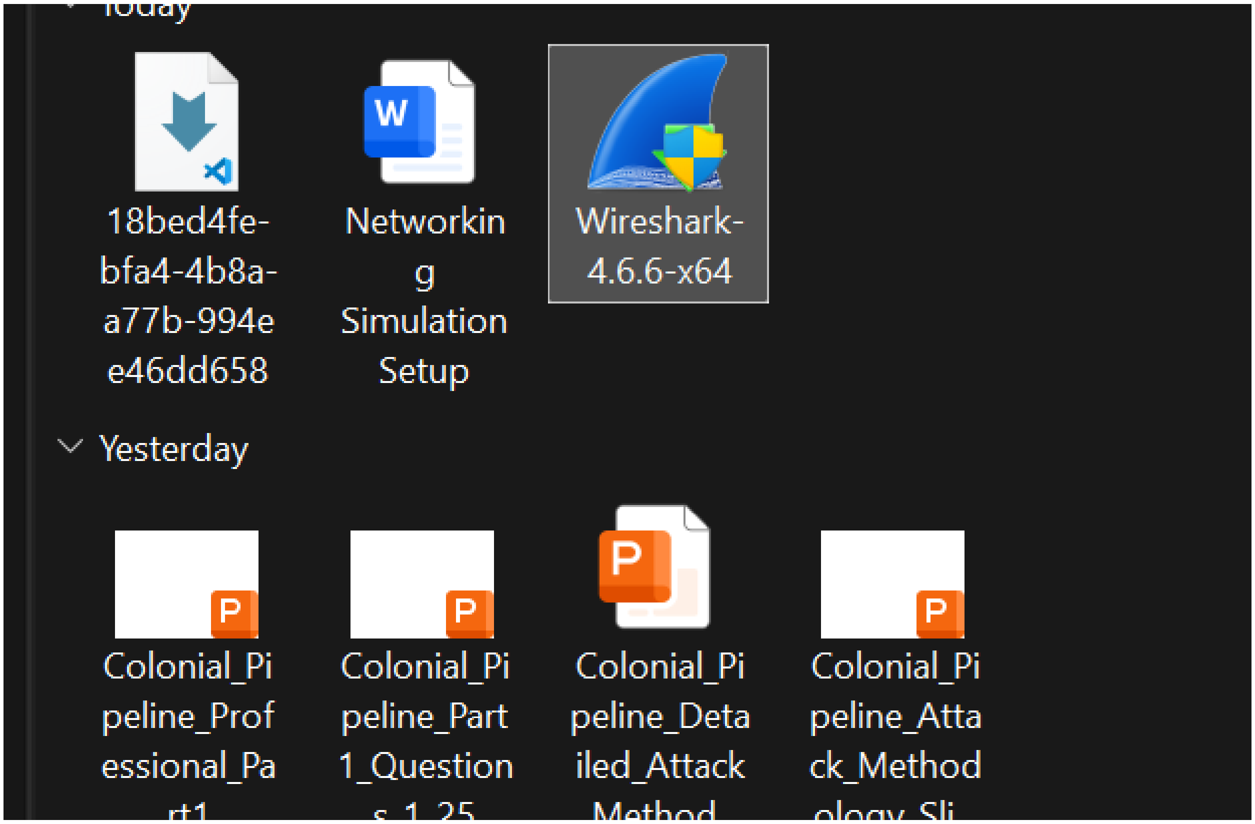
Installation Steps:

1. Downloaded Wireshark from the official website.
2. Executed the installer package.
3. Accepted the license agreement.
4. Installed Npcap packet capture drivers.
5. Completed software installation.
6. Verified successful launch.

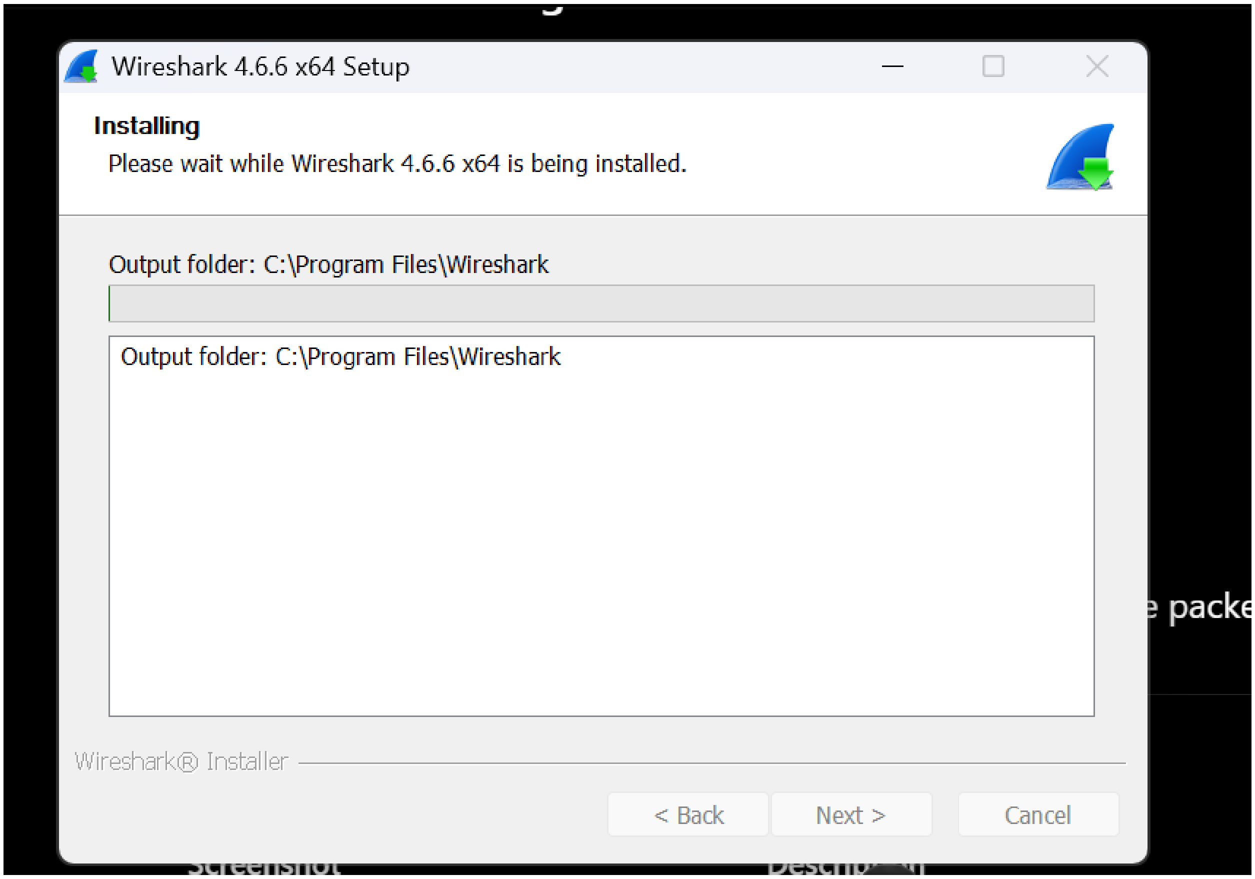
Wireshark Download Page



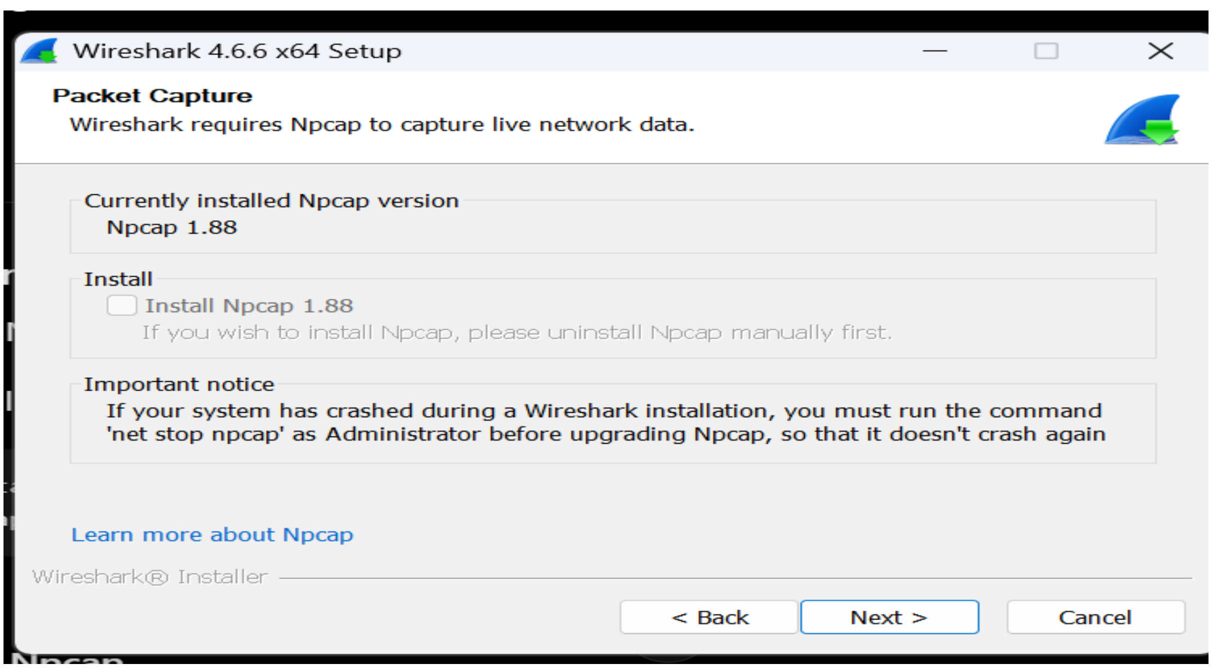
Downloaded Wireshark Installer



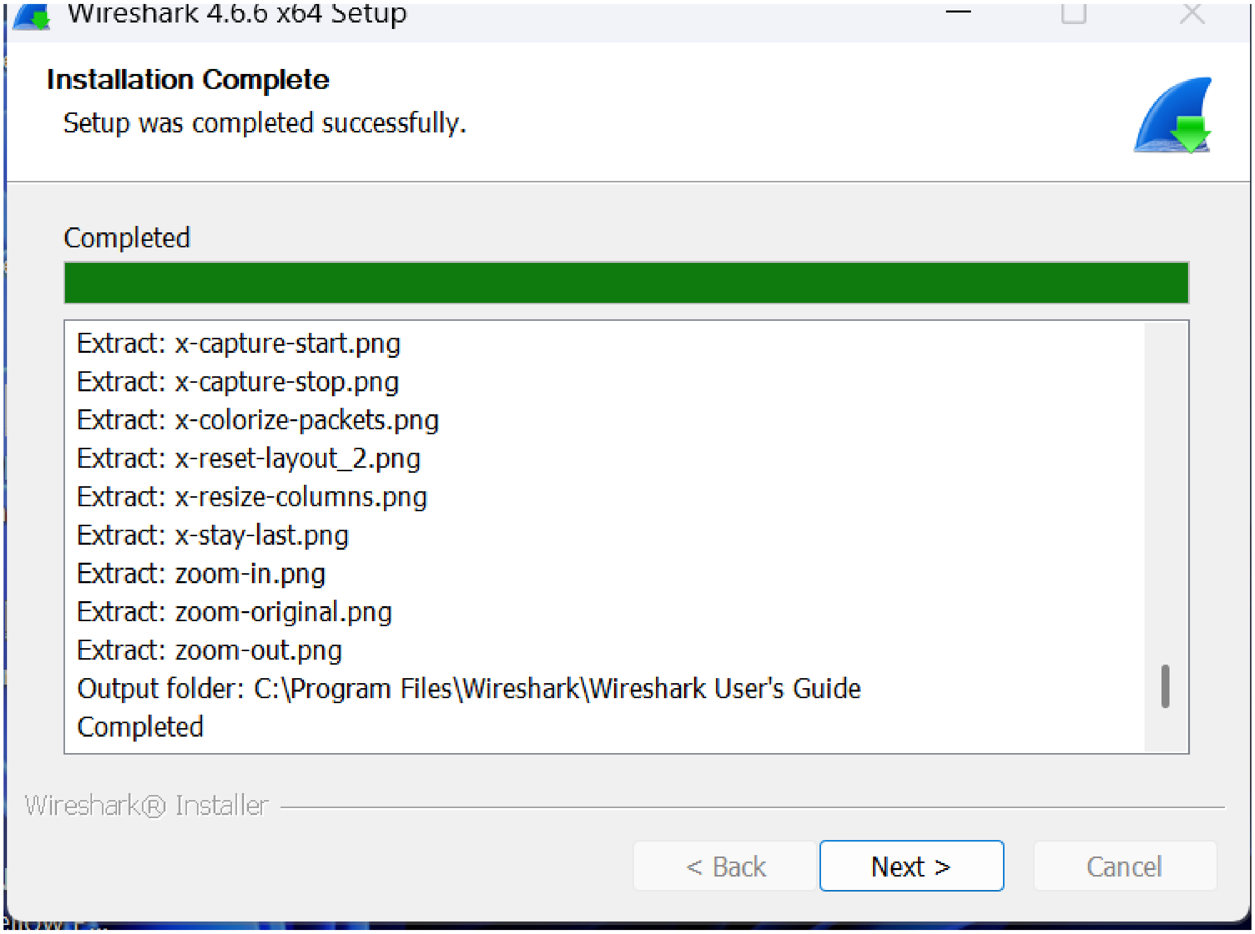
Wireshark Installation Wizard



Npcap Installation



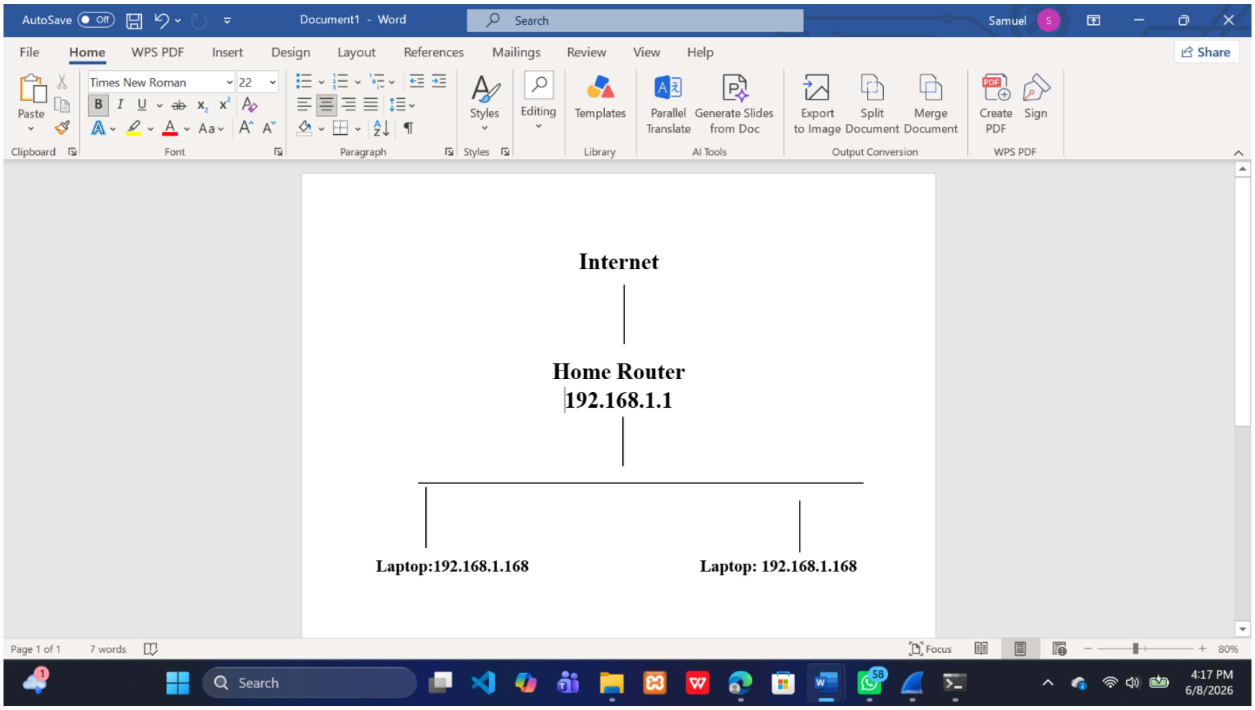
Installation Completed Successfully



8. NETWORK TOPOLOGY AND ARCHITECTURE

The monitoring environment consisted of a wireless home network.

Figure 1: Home Network Topology



The router acted as the default gateway and provided network connectivity to all devices.

Insert Figure 1: Home Network Topology Diagram

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9. NETWORK DEVICE CONFIGURATION

Table 2: Connected Devices

Device: Home Router IP Address: 192.168.1.1 Function: Default Gateway

Device: Laptop IP Address: 192.168.1.168 Function: Packet Analysis Workstation

Device: Android Phone IP Address: 192.168.1.136 Function: Network Client Device

Network Configuration:

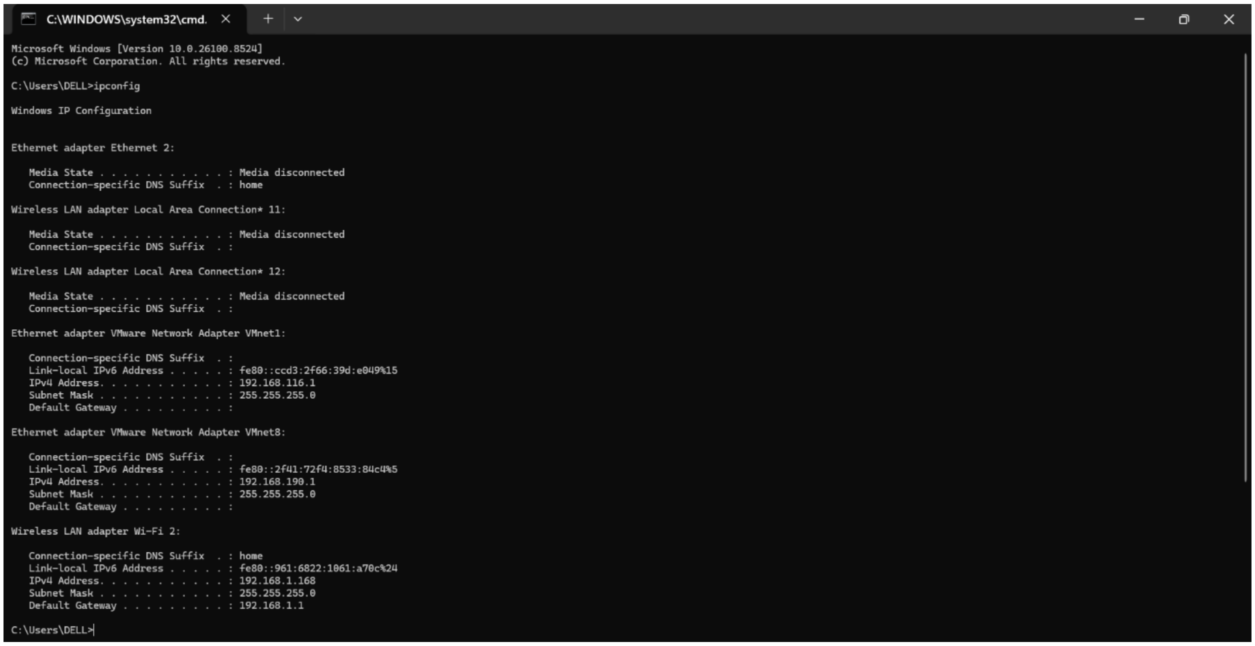
IPv4 Network: 192.168.1.0/24

Subnet Mask: 255.255.255.0

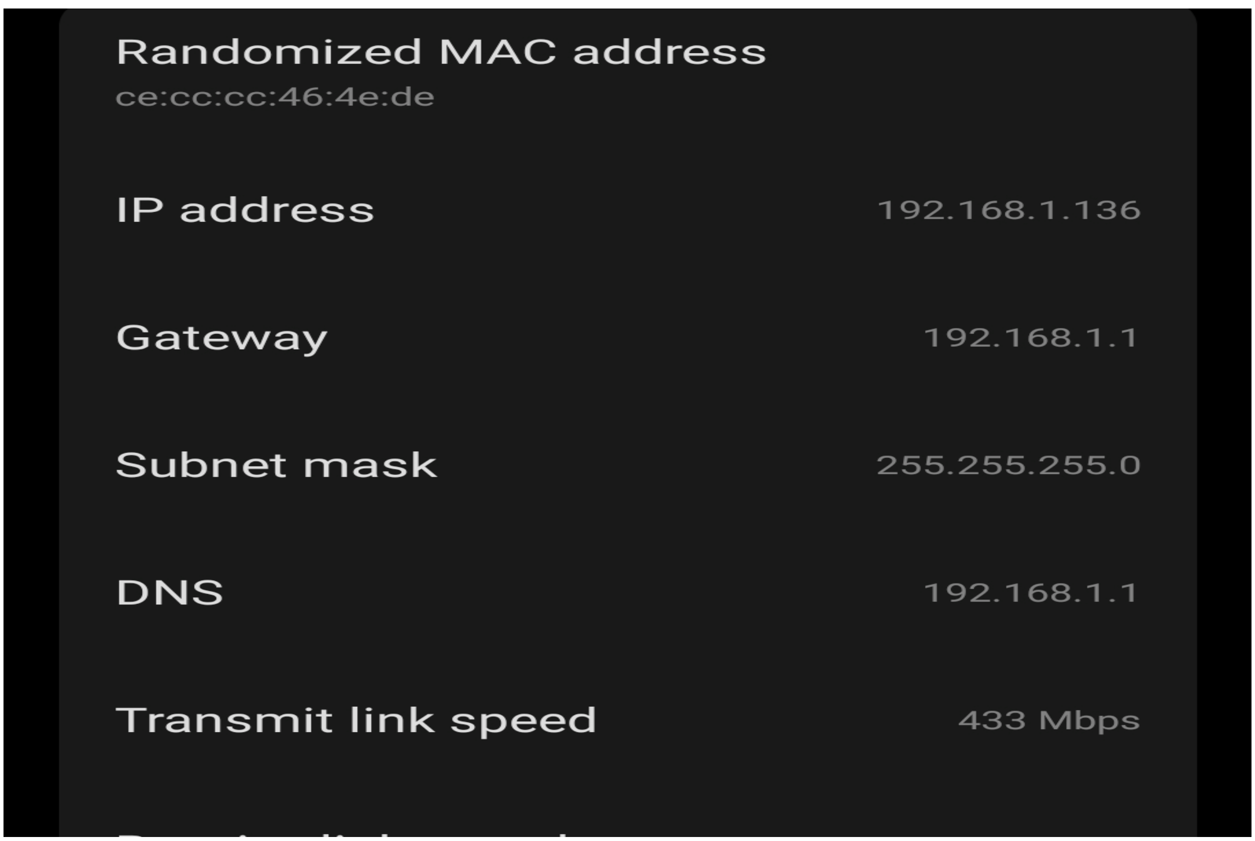
Default Gateway: 192.168.1.1

DNS Server: 192.168.1.1

Laptop IP Configuration (ipconfig)



Android Network Configuration



10.CONNECTIVITY TESTING

Connectivity testing was performed to verify communication between devices.

Command Used:

ping 192.168.1.1

Results:

Packets Sent = 4

Packets Received = 4

Packet Loss = 0%

Average Response Time = 0ms

Interpretation:

The successful ping responses confirmed that the laptop could communicate successfully with the router.

Ping Connectivity Test

```
C:\Users\DELL>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\DELL>
```

11.ARP TABLE ANALYSIS

Address Resolution Protocol (ARP) was used to identify devices communicating on the local network.

Command Used:

arp -a

Purpose:

• Identify local devices • Map IP addresses to MAC addresses • Verify local communications

Findings:

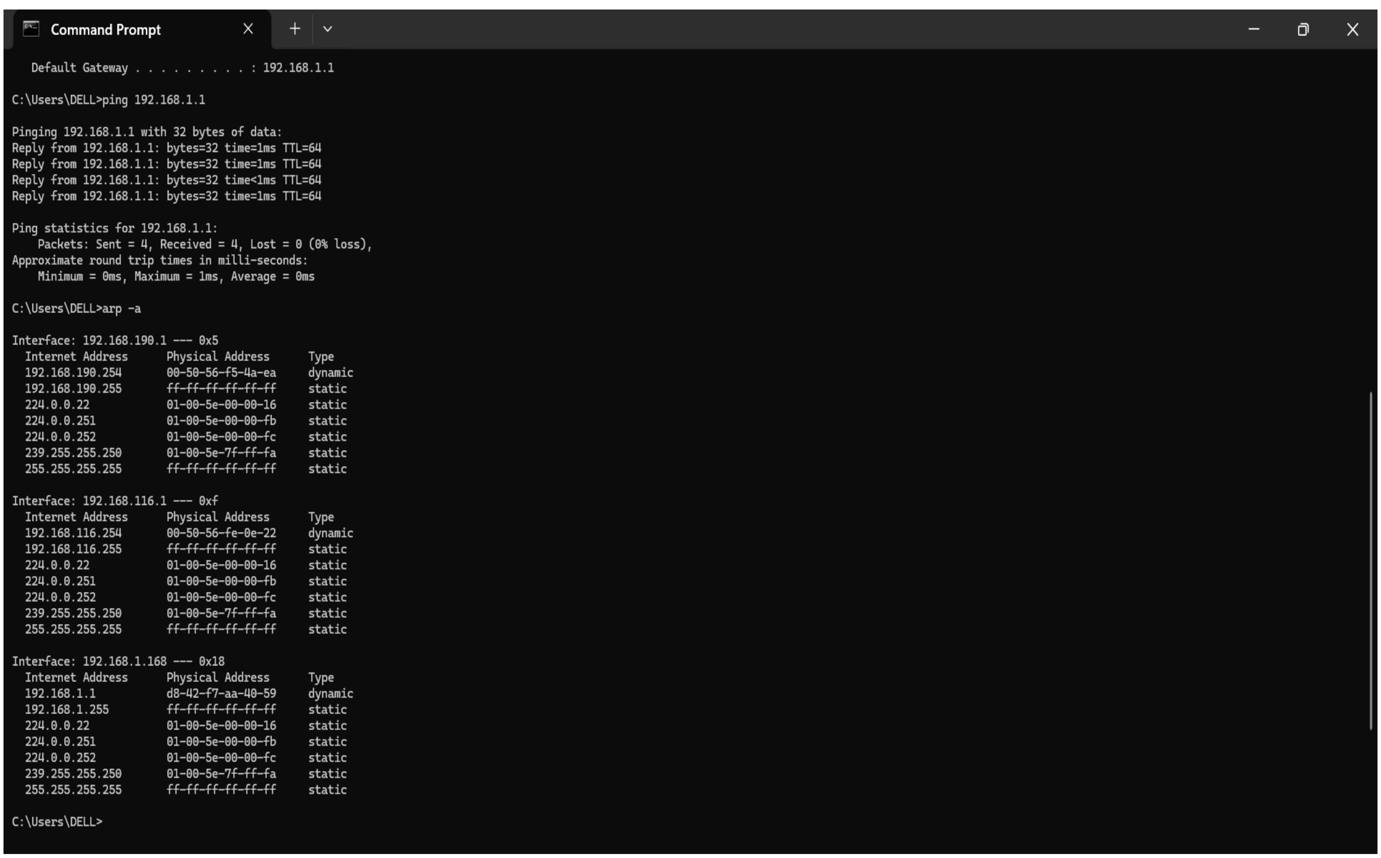
• Router detected successfully • Network communications verified • Device addressing confirmed

Cybersecurity Importance:

ARP analysis assists in:

• Device discovery • Network troubleshooting • Detection of unauthorized devices • Security investigations

ARP Table Output



12.SECURITY CONTROLS IMPLEMENTATION

12.1 Windows Defender Firewall

Windows Defender Firewall was enabled on the system.

Observed Configuration:

Firewall State: Enabled

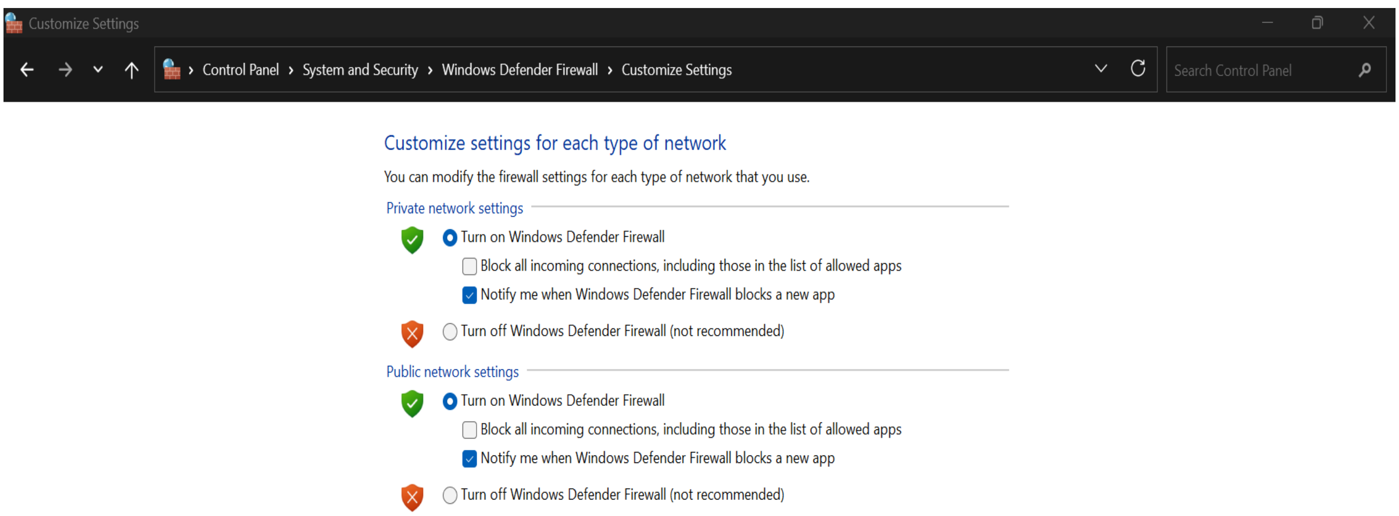
Public Network Protection: Enabled

Inbound Connections: Restricted

Benefits:

• Blocks unauthorized access • Monitors network traffic • Reduces attack surface • Protects against malicious connections

Windows Defender Firewall Enabled



12.2 Wireless Security Assessment

Wireless security settings were examined.

Observed Security Configuration:

Security Type: WPA2-Personal

Encryption: AES

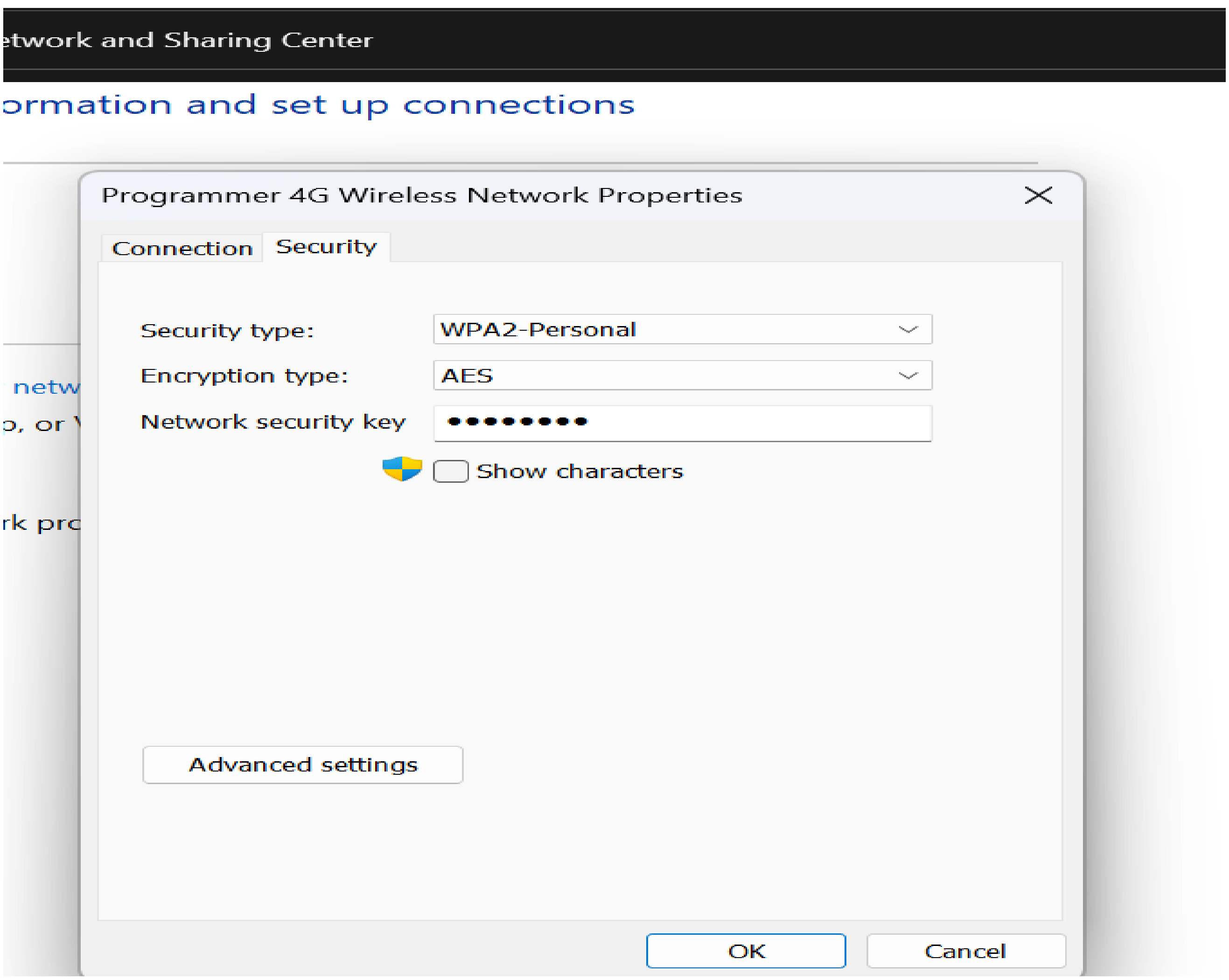
Analysis:

WPA2 with AES encryption protects wireless communications against unauthorized access and interception.

Benefits:

• Data confidentiality • Network protection • Secure wireless communications

Wireless Security Configuration



13.NETWORK INTERFACE DETECTION

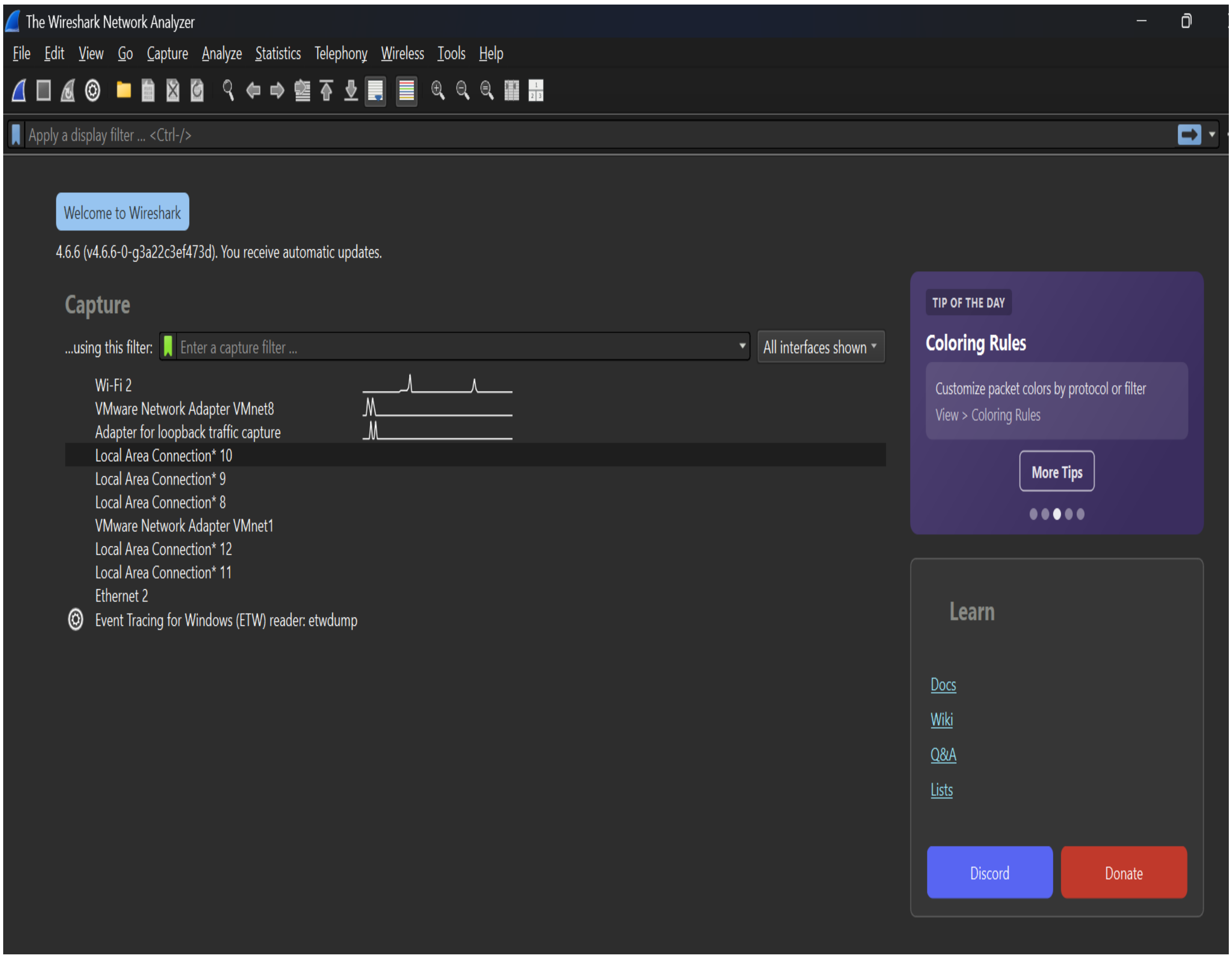
After launching Wireshark, several network interfaces were detected.

The active interface selected for packet capture was:

Wi-Fi 2

This interface showed active network traffic and was selected for analysis.

Available Network Interfaces



14.LIVE PACKET CAPTURE

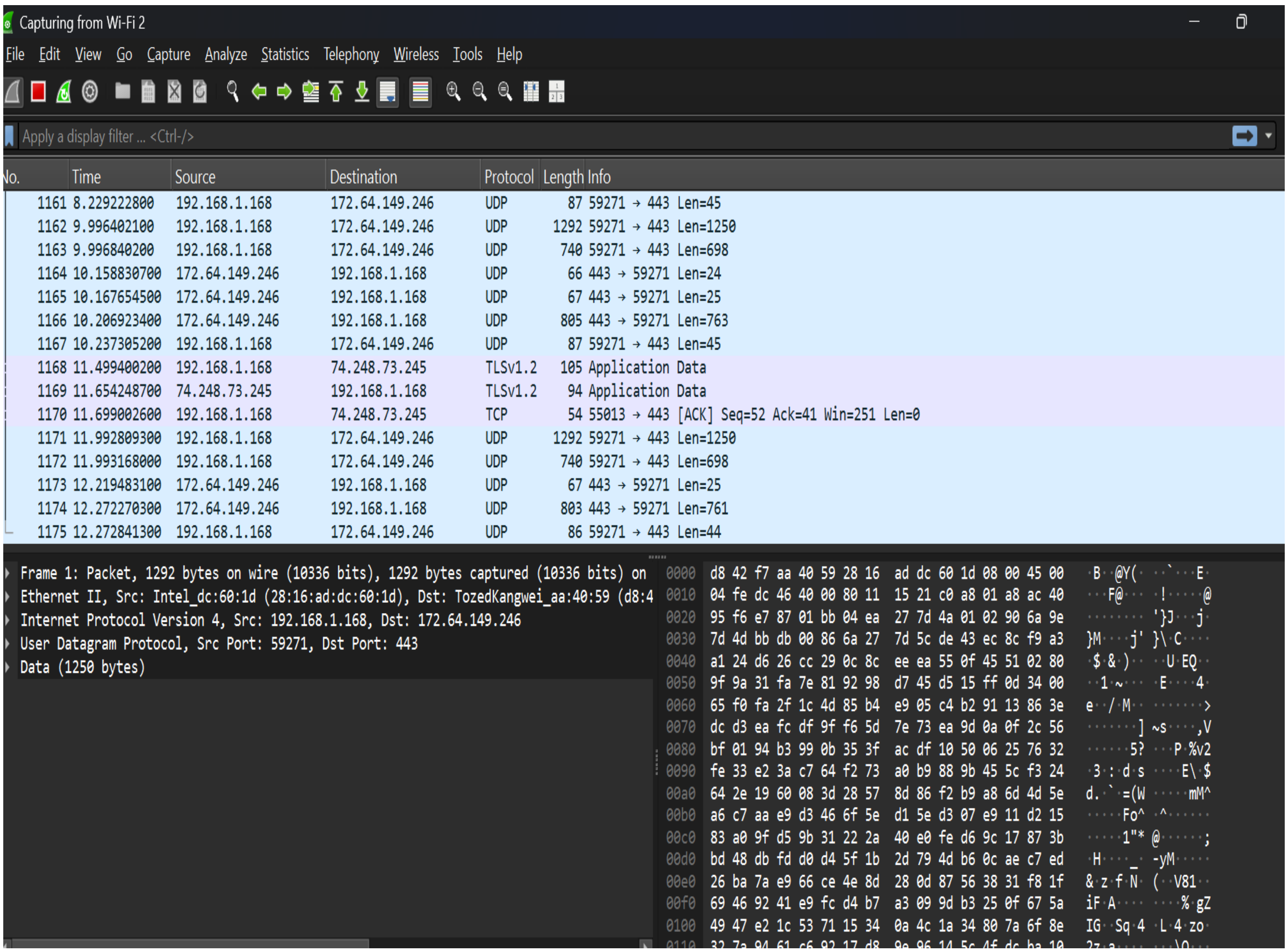
A live packet capture session was initiated on the Wi-Fi 2 adapter.

Traffic was generated through:

• Web browsing • DNS lookups • HTTPS communications • Background network activities

Wireshark successfully captured and displayed packets in real time.

Live Packet Capture



15.PROTOCOL ANALYSIS

15.1 DNS Analysis

Protocol: DNS (Domain Name System)

Purpose:

DNS converts domain names into IP addresses.

Observed Activities:

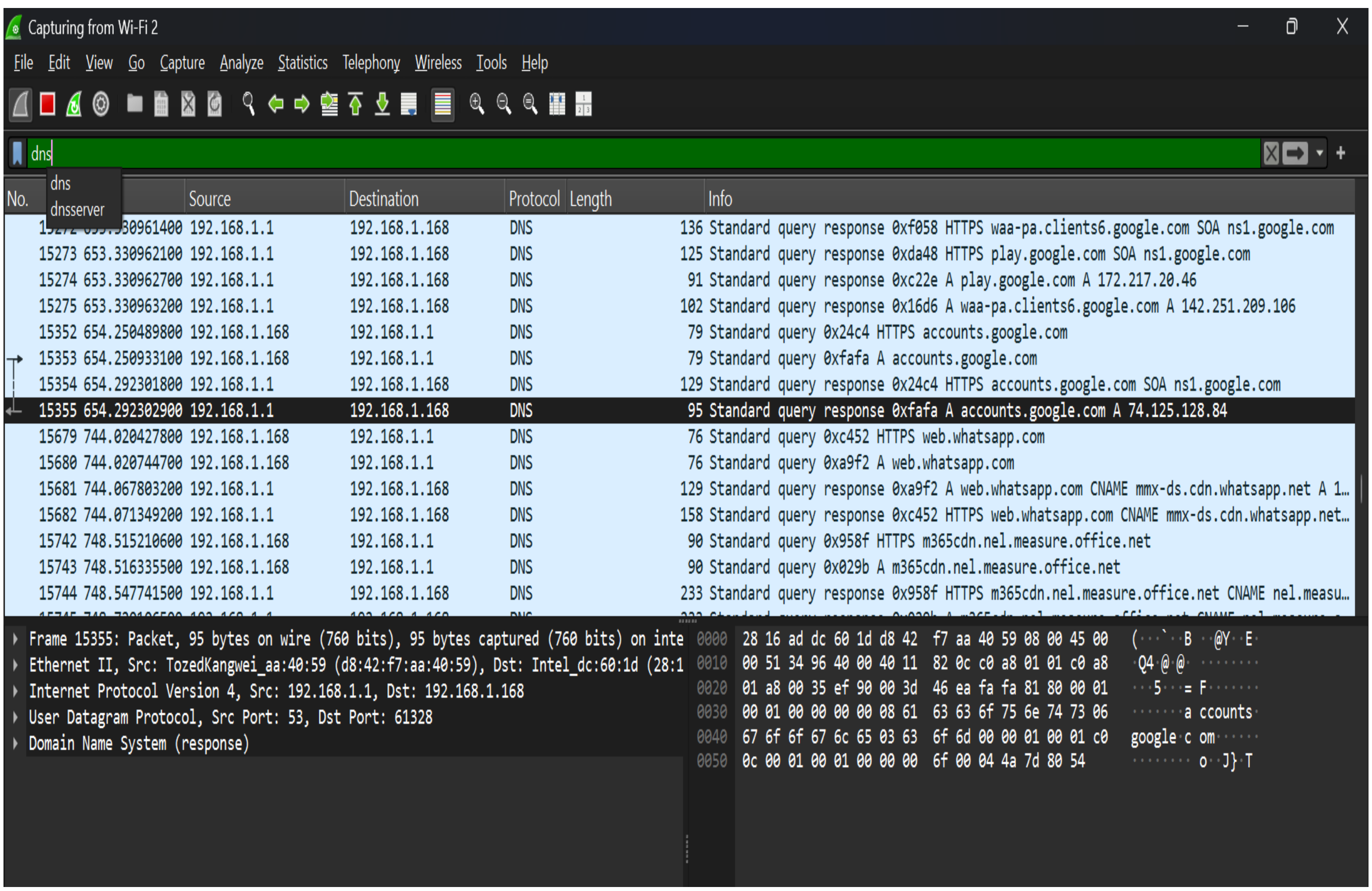
• DNS Queries • DNS Responses • Name Resolution

Cybersecurity Relevance:

DNS monitoring helps identify:

• Malicious domains • Command-and-Control servers • DNS tunneling attacks

DNS Protocol Analysis



15.2 ICMP Analysis

Protocol: ICMP (Internet Control Message Protocol)

Purpose:

Used for:

• Ping Requests • Ping Replies • Network Diagnostics

Observed Activities:

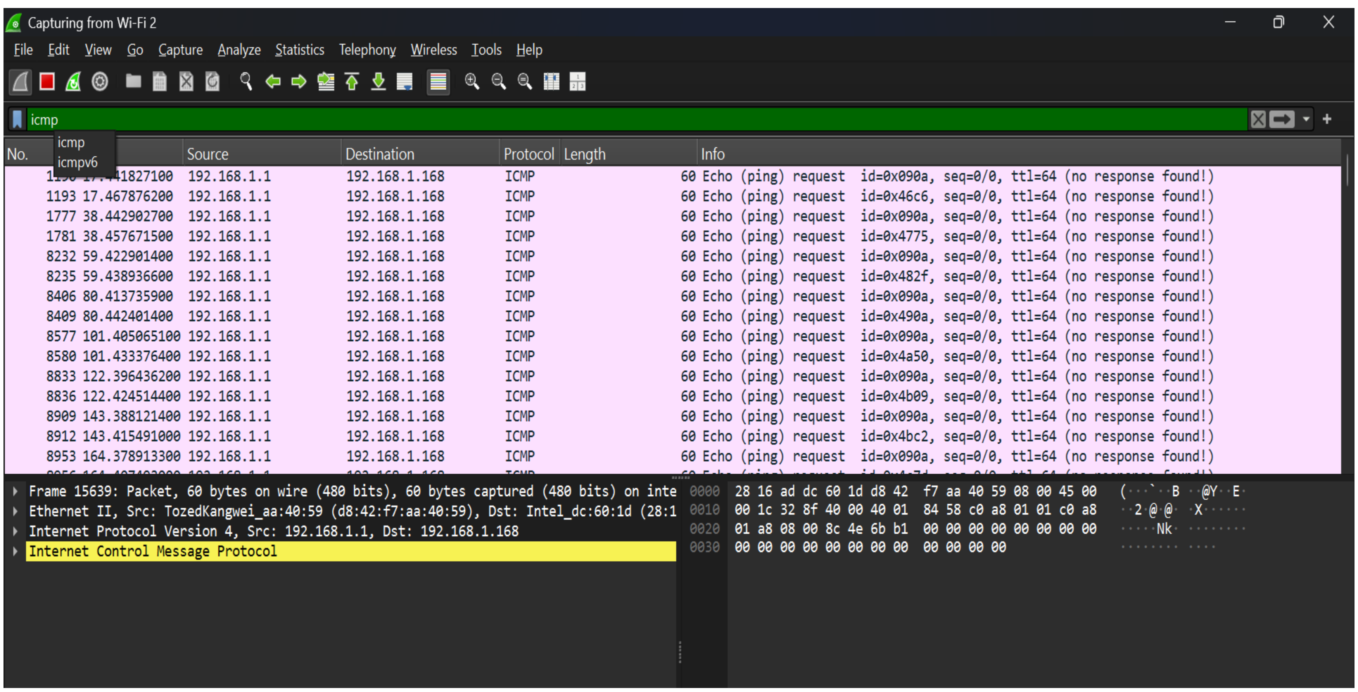
• Echo Requests • Echo Responses

Cybersecurity Relevance:

ICMP is useful for:

• Connectivity testing • Host discovery • Troubleshooting

ICMP Protocol Analysis



15.3 TCP Analysis

Protocol: TCP (Transmission Control Protocol)

Purpose:

Provides reliable communication between devices.

Observed Activities:

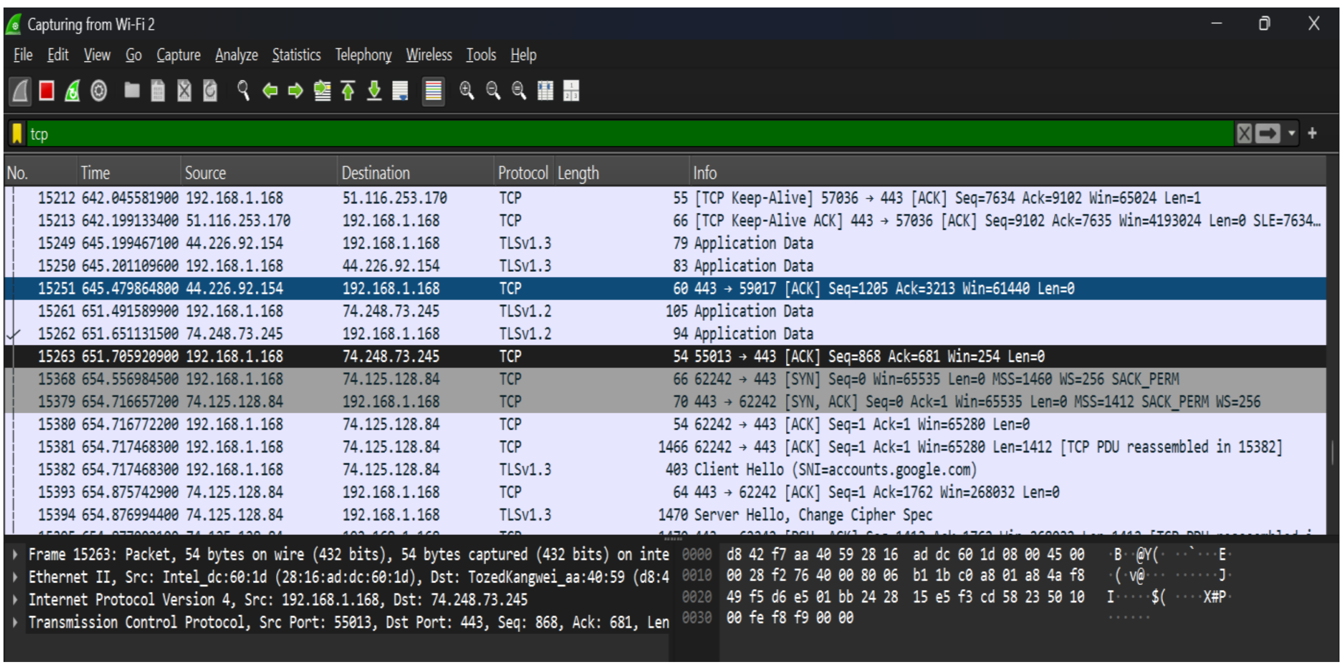
• TCP Handshake • ACK Packets • Established Connections

Cybersecurity Relevance:

TCP analysis assists in:

• Monitoring connections • Detecting anomalies • Investigating suspicious traffic

TCP Protocol Analysis



16.SECURITY FINDINGS

The following protocols were observed:

• DNS • TCP • ICMP • UDP • TLS

The captured traffic demonstrated normal network communications.

No suspicious traffic was observed during the monitoring period.

Implemented Security Controls:

✓ Windows Defender Firewall Enabled

✓ WPA2 Wireless Security Enabled

✓ AES Encryption Enabled

✓ Wireshark Monitoring Implemented

17.TESTING AND VALIDATION

Validation Activities Completed:

✓ Wireshark Installed Successfully

✓ Npcap Installed Successfully

✓ Interface Detection Successful

✓ Live Traffic Capture Successful

✓ DNS Analysis Successful

✓ ICMP Analysis Successful

✓ TCP Analysis Successful

✓ Connectivity Testing Successful

✓ Firewall Verification Successful

✓ Wireless Security Verification Successful

18.CONCLUSION

This project successfully established a practical cybersecurity monitoring environment using Wireshark and Npcap. Live network traffic was captured and analyzed through the Wi-Fi 2 interface, enabling observation of real-world network communications.

The project demonstrated the ability to identify and analyze key network protocols including DNS, ICMP, TCP, UDP, and TLS. Network connectivity testing confirmed stable communication between devices, while ARP analysis provided visibility into local network addressing.

Security controls including Windows Defender Firewall and WPA2 wireless encryption were verified and documented. These controls contribute to protecting systems against unauthorized access and network-based threats.

The project provided valuable practical experience in packet capture, protocol analysis, network troubleshooting, security monitoring, and traffic inspection. These skills are essential for cybersecurity analysts, network administrators, SOC analysts, digital forensic investigators, and penetration testers responsible for protecting modern networks.

The objectives of the project were successfully achieved, and the implementation demonstrated the practical application of industry-standard tools for cybersecurity monitoring and network analysis.